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Platform wheel dent

Abstract

A stopping station (15) for use in a transportation network (100) is disclosed. The stopping station (15) comprises a passenger boarding/alighting platform (30) and a plurality of vehicle stopping surfaces (90). The passenger boarding/alighting platform (30) has an upper surface (80) for boarding of passengers into a vehicle (20). The passenger boarding/alighting platform (30) also has a vehicle-facing surface (85) comprising at least two of a cutout (40) for docking of a wheel (45) of the vehicle to the passenger boarding/alighting platform (30). The vehicle stopping surface (90) is adjacent to the vehicle-facing surface (85) and is used for stopping of the vehicle (20) during the boarding/alighting of the passengers in/from the vehicle (20).

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Background/Summary

FIELD OF THE INVENTION

(1) This application claims priority of the British Patent Application number GB2103275.0, filed on 10 Mar. 2021. The entire disclosure of the British Patent Application number GB2103275.0 is hereby incorporated herein by reference

(2) The field of the invention relates to a passenger boarding/alighting platform having cutouts for stopping of a vehicle close to the passenger boarding/alighting platform and methods for the controlling the movement of the vehicle at the passenger boarding/alighting platform having the cutouts.

BACKGROUND OF THE INVENTION

(3) Transportation networks for transportation of passengers between stopping stations often comprise platforms for boarding/alighting of passengers in/from a vehicle. Depending on the type of platform and the type of vehicle used in the transportation network, a gap between the vehicle and the platform is commonly found in today's transportation networks for allowing the vehicle to approach the stopping station before boarding/alighting passengers or for leaving the stopping station after boarding/alighting of the passengers at the stopping station. The passengers have to cross this gap between the vehicle and the stopping station in order to board/alight in/from the vehicle. Crossing this gap can lead to accidents and/or be a disadvantage for disabled passengers. Different solutions for reducing the gap between the vehicle and the platform are known and used in the current transportation networks around the globe, such as extendible or movable platforms, or “kneeling” vehicles.

(4) MTI Report 12-31 “Automated Transit Networks (ATN): A Review of the State of the Industry and Prospects for the Future, MTI Report 1227”. Burford Furman et al. Mineta Transportation Institute Publications (2014) discloses different types of stopping stations as currently used in transportation networks. Multiple concepts for stopping vehicles at the side of a road are disclosed. A system for reducing a gap between the vehicle and the platform is not, however, disclosed in the report.

(5) Self-driving cars or vehicles may be used in the transportation networks. Self-driving cars are often described as being “autonomous”, but in practice, there are different classes or levels of vehicle autonomy. The degree of vehicle autonomy is typically divided in five levels, as set out by the On-Road Automated Driving (ORAD) committee of the Society of Automotive Engineers (SAE) in “Taxonomy and definitions for terms related to driving automation systems for on-road motor vehicles” published in Recommended Practice SAE J 3016 on 15 Jun. 2018. Level 0 refers to a vehicle that has no driving automation. The driver of the vehicle is fully in charge of operating the movement of the vehicle. Vehicles of Level 0 may include safety systems such as, for example, a collision avoidance alert. Level 1 refers to vehicles having at least one driving assistance feature such as an acceleration or braking assist system. The driver is responsible for the driving tasks but is supported by the driving assist system which is capable of affecting the movement of the vehicle. Level 2 describes vehicles having more than one assist system for actively affecting the movement of the vehicle. The driver, in Level 2, is still responsible for the driving tasks and must actively monitor the trajectory of the vehicle at all times. The driver is, however, actively supported by the assist systems. Level 3 describes a so-called “conditional automation” of the vehicle. The vehicle is capable of autonomously driving in certain situations and with limitations. The driver is not required to actively monitor the assist system but is, however, required to take control of a driving situation if requested by the assist system. Level 4 describes autonomously travelling vehicles

which are capable of travelling specific routes under normal conditions without human supervision. The vehicles of Level 4 can therefore operate without a driver but might need remote human supervision in case of conflict situations, travelling in remote areas, or when travelling extreme weather conditions. Level 5 Automation describes fully autonomously driving vehicles. No human interaction is required at any time for the operation of the vehicles.

(6) Korean patent application KR 2003/0084874 A (Eom Gi Yeong et al, assigned to KRRI) discloses a safety device installed on a platform to reduce a gap between a vehicle and the platform. The safety device comprises an elastic member that is connected to a fixing member attached to the platform. The elastic member contacts a door of the vehicle when the vehicle is in the station. This contacting of the door of the vehicle helps to close the gap between the platform and the vehicle. The elastic member for the safety device has different degrees of flexibility in a horizontal direction and a vertical direction. The elastic member is flexible in the horizontal direction and absorbs horizontal forces from the vehicle entering the station by deforming the elastic member. The elastic member is not, however, flexible in the vertical direction and is able to withstand a force applied by a passenger standing on the flexible member. A use of cutouts in the platform for the reducing the gap between the vehicle and the platform is not, however, disclosed in the Korean patent application.

(7) Japanese patent application JP H11-209926 A (Tanaka, assigned to Hino Auto Body Ltd) discloses a stopping platform for a vehicle that is formed by notching a sidewalk, wherein the sidewalk is a surface that is a step higher from the road. The sidewalk is notched to fit the length of one vehicle. The notched part is approximately formed to be in the shape of a triangle. Edging stones are arranged at the edge of the sidewalk to distinguish the platform from the road. The Japanese patent application discloses the notching of the sidewalk to accommodate the vehicle during the boarding/alighting of the vehicle. Specific cutouts for the wheels of the vehicle are not, however, disclosed in the patent application.

(8) The prior art discloses solutions for reducing the gap between the vehicle and the platform and rely on reduction of the size of the gap by separate devices that are attached to the platform or the platform is notched to match the size of the vehicles. The prior art does not, however, disclose a system or method for reducing the gap between the vehicle by allowing the vehicle to stop in immediate proximity to the platform.

SUMMARY OF THE INVENTION

(9) The present document describes a stopping station for use in a transportation network. The stopping station comprises a passenger boarding/alighting platform and a vehicle stopping surface adjacent to the passenger boarding/alighting platform. The passenger boarding/alighting platform comprises an upper surface for boarding of passengers in a vehicle and a vehicle-facing surface. The vehicle-facing surface comprises at least two cutouts for docking a wheel of the vehicle to the cutouts of the passenger boarding/alighting platform. The vehicle stopping surface is used for stopping of the vehicle during boarding/alighting of the passengers in/from the vehicle. The form of the cutout is dimensioned according to a wheel hull curve of the wheel of the vehicle and enables rotation of the wheel when the wheel is docked to the cutout. The number of cutouts on a side of the vehicle facing the passenger boarding/alighting platform correlates to the number of wheels of the vehicle. The stopping station of the present document allows, due to the docking of the wheels to the cutouts of the platform, a reduction in the size of the gap between the vehicle and the platform.

(10) The present document further describes a method for the boarding/alighting of the passengers in/from the vehicle at the passenger boarding/alighting platform. The method comprises approaching of the vehicle to the passenger boarding/alighting platform and aligning the wheels of the vehicle with a plurality of cutouts disposed in the vehicle-facing surface of the passenger boarding/alighting platform. The vehicle is caused to stop proximate to the passenger boarding/alighting platform and the wheels are docked to the cutouts of the passenger

boarding/alighting platform. The passengers board/alight the vehicle in/from the vehicle and the vehicle subsequently leaves the stopping station by the vehicle. In one aspect of the method, the vehicle detects the stopping station using an onboard positioning unit and an onboard processor of the vehicle.

(11) The present document further describes a method for management of vehicles at the stopping station in the transportation network.

Description

DESCRIPTION OF THE FIGURES

(1) FIG. 1 shows a top view of a stopping station comprising a cutout.

(2) FIG. 2 shows a perspective view of the stopping station comprising the cutout.

(3) FIGS. 3A, 3B, 3C, and 3D show a perspective view of a vehicle stopping at the stopping station.

(4) FIG. 4 shows a flowchart describing a method for controlling a movement of vehicles on a selected one of a plurality of vehicle stopping surfaces at a stopping station in the transportation network.

(5) FIG. 5 shows a flowchart describing a method for controlling a departure of a vehicle requesting to depart from a stopping station.

(6) FIG. 6 shows a flowchart describing a method for boarding/alighting of a passenger at a boarding/alighting platform.

(7) FIG. 7 shows a flowchart describing a method for management of vehicles at the boarding/alighting platform.

DETAILED DESCRIPTION OF THE INVENTION

(8) The invention will now be described on the basis of the figures. It will be understood that the embodiments and aspects of the invention described herein are only examples and do not limit the protective scope of the claims in any way. The invention is defined by the claims and their equivalents. It will be understood that features of one aspect or embodiment of the invention can be combined with a feature of a different aspect or aspects and/or embodiments of the invention.

(9) FIG. 1 shows a top view of a stopping station **15** for use in a transportation network **100** with a plurality of autonomous vehicles **20**. The stopping station **15** comprises a passenger boarding/alighting platform **30** and a plurality of vehicle stopping surfaces **90** at which the autonomous vehicles **20** can stop to enable boarding and/or alighting of passengers. The plurality of the vehicle stopping surfaces **90** are, for example, arranged adjacent to each other at the stopping station **15**. The stopping station **15** further comprises a stopping station control unit **16** and a stopping station communication unit **17**. The stopping station control unit **16** knows the position of the vehicle stopping surfaces **90** within the stopping station **15** and is capable of identifying the individual ones of the vehicle stopping surfaces **90** and the stopping station communications unit **17** communicates with ones of the autonomous vehicles **20** at or near the stopping station **15**.

(10) The passenger boarding/alighting platform **30** has an upper surface **80** and a vehicle-facing surface **85**. The upper surface **80** is for boarding/alighting of passengers (not shown) into a vehicle **20**. The vehicle-facing surface **85** is adjacent to the plurality of the vehicle stopping surfaces **90** and comprises at least two of a cutout **40** for docking of a wheel **45** of the vehicle **20**. The transportation network **100** comprises a plurality of stopping stations **15** disposed in the transportation network **100** and connected by roads or tracks (not shown). The transportation network **100** comprises at least one vehicle **20** for travelling between the stopping stations **15** by a passenger (not shown).

(11) The transportation network further comprises at least one roadside control unit (not shown) for controlling of the vehicles **20**. The roadside control unit is connected to the stopping station control

unit **16** and capable of communicating with the vehicle **20**. The roadside control unit is used for controlling of the vehicles **20** when travelling between the stopping stations **15** in the transportation network **100**. The roadside control unit will inform the stopping station control unit **16** of approaching ones of the vehicles **20**. The stopping station control unit **16** will take over from the roadside control unit a management of the vehicles **20** during approaching and departing of the vehicles **20** to/from the stopping station **15** (as is further elaborated in the description of FIGS. **4** to **7**).

(12) The vehicle **20** has a front and a rear. The vehicle **20** has a side **21** and a plurality of wheels **45**. The wheels **45** are located, for example, near the front and the rear of the vehicle **20**. The wheels **45** located at the front of the vehicle **20** are termed front wheels **45f** and the wheels **45** located at the rear are termed rear wheels **45r**. The wheels **45** can be flush with the side **21** of the vehicle **20**, stick out from the side **21** of the vehicle **20**, or can be shifted to be underneath the inside of the vehicle **20**. This shifting of the wheels **45** to the inside may, however, have an impact on a stability of the vehicle **20** during maneuvering in curves.

(13) The wheels **45** define a wheel hull curve **50** (the wheel hull curve **50** is only shown on one wheel **45** in FIG. **1** for the sake of clarity of the drawing). The wheel hull curve **50** describes a geometrical outline or surface created by possible positions of an outer surface of the wheel **45** when the wheel **45** rotates and moves during the steering of the wheel **45**. More precisely, the wheel hull curve **50** describes, as it is used here, the sum of possible positions of the outer surface of the wheel **45** that the wheel **45** might reach during normal steering operation and during normal driving operation. The wheel hull curves **50** of the wheels **45** of the vehicle **20** might be different and depend on, for example, the position of the wheel **45** relative to the vehicle **20**.

(14) The vehicle **20** comprises an onboard processor **27**, a vehicle memory **28**, a vehicle antenna **25**, and a vehicle positioning unit **26**. The onboard processor **27** is used for calculating a route to the stopping station **15**, and on arrival at the stopping station **15**, the onboard processor **27** can also be used to calculate the route to a vehicle stopping surface **90** within the stopping station **15**. The vehicle memory **28** is used for storing a structure model **75**. The structure model **75** comprises items of data on the transportation network **100** and a position of the stopping stations **15** in the transportation network **100**. The vehicle memory **28** is also used for storing further items of data on, for example, the route of the vehicle **20**. The vehicle antenna **25** is used for communicating, for example, with beacons disposed in the transportation network **100** or, for example, with the stopping station control unit **16** of the stopping station **15**. The onboard positioning unit **26** is used for detecting (see also step **S100** described in FIG. **6**) the stopping station **15** in the transportation network **100**. The vehicle **20** also includes assist systems of Level 2 or Level 3, as described above. The vehicle **20** is, in one aspect, capable of autonomously travelling in the transportation network **100**, using the structure model **75** and the onboard processor **27**. The vehicle **20** does therefore not require a driver for driving of the vehicle **20**.

(15) The wheels **45** can be steered individually or in groups by the onboard processor **27**. A first group for the steering of the wheels **45** are the front wheels **45f** and a second group for the steering are the rear wheels **45r**. The steering of the wheels **45** comprises rotating of the wheels **45** around a pivot point **PP** (the pivot point **PP** is only shown on one wheel **45** in FIG. **1** for the sake of clarity of the drawing). This rotating of the wheel **45** for the steering of the wheel **45** comprises changing an orientation of the wheel **45** relative to the direction of a longitudinal center axis **AX1** of the vehicle **20** around the pivot point **PP**. The pivot point **PP** of the wheel **45** is, when seen from above (as depicted in FIG. **1**), located inside of the wheel hull curve **50**, inside of the wheel **45**, or inside of the vehicle **20**.

(16) The wheels **45** are driven individually or in groups by one or several motors (not shown) for powering the vehicle **20**. For example, one group are the front wheels **45f** and another group are the rear wheels **45r**. In a further aspect, only one of the groups of the front wheels **45f** or the rear wheels **45r** is powered by the motor. The other one of the groups of the front wheels **45f** or the rear

wheels **45r** is not powered by the motor.

(17) The wheels **45** can be attached to the vehicle **20** at a camber angle for improving stability of the vehicle **20** during maneuvering or steering around a curve. The camber angle describes an angle between a vertical axis of the vehicle **20** and a vertical axis of the wheel **45**, as viewed from the front of the vehicle **20**. A negative camber angle (upper portion of the wheels **45** leaning inward/towards the center axis **AX1** of the vehicle **20**) can, for example, improve the stability of the vehicle when maneuvering through a sharp curve.

(18) The number of cutouts **40** disposed on the passenger boarding/alighting platform **30** correlates to the number of the wheels **45** on one side of the vehicle **20**. The cutout **40** is dimensioned according to the wheel hull curve **50** of at least one wheel **45** of the vehicle **20** and enables rotation of the docked wheel **45** within the cutout **40**. A gap **60** between the vehicle **20** and the passenger boarding/alighting platform **30** of the stopping station **15** can be reduced by enabling the wheel **45** to rotate in the docked position. The cutout **40** is dimensioned to have a cutout width **40w**, a cutout height **40h**, and a cutout depth **40d**. The cutout width **40w**, the cutout height **40h**, and the cutout depth **40d** are dimensioned based on the wheel hull curve **50**.

(19) The cutout **40** can, in a first aspect, be dimensioned to be similar to the wheel hull curve **50** in shape and size. The wheels **45** will dock closely to the cutout **40** in this first aspect. The cutout **40** can, in a second aspect, also be dimensioned to be similar in shape to the wheel hull curve **50** but to be bigger in size than the wheel hull curve **50**. The wheels **45** will dock to the cutout **40** with an offset in this second aspect. The cutout **40** could, in a third aspect, also be dimensioned to be different in shape and/or size from the wheel hull curve **50**. The cutout **40** will be dimensioned to accommodate at least the wheel hull curve **50** in this third aspect. The gap **60** is reduced in the first to the third case by allowing the wheel **45** to rotate in the cutout **40**.

(20) FIG. 2 shows a perspective view of the stopping station **15** in the transportation network **100**. FIG. 2 shows the stopping station **15** comprising cutouts **40** extending from the upper surface **80** of the passenger boarding/alighting platform **30** to the plurality of the vehicle stopping surfaces **90** adjacent to the passenger boarding/alighting platform **30**. The cutout height **40w** is the same as a vertical distance (step) between the upper surface **80** and the plurality of the vehicle stopping surfaces **90**.

(21) FIGS. 3A, 3B, 3C, and 3D show different examples of the cutouts **40** for the docking of the wheels **45** to the passenger boarding/alighting platform **30**. The vehicle **20** approaches (step **S120** described in FIG. 6) the passenger boarding/alighting platform **30** by driving onto one of the plurality of the vehicle stopping surfaces **90**. A driving path of the vehicle **20** during the approaching of the vehicle **20** to the passenger boarding/alighting platform **30** depends on, for example, the number of wheels **45** being steered by the onboard processor **27**.

(22) The cutouts **40** are dimensioned according to the wheel hull curve **50** as shown in FIG. 3A. In this first example, the wheel **45** rotates around the pivot point **PP** within the wheel hull curve **50**. The cutout width **40w**, the cutout height **40h**, and the cutout depth **40d** are dimensioned to allow the wheel **45** to rotate around the pivot point **PP** during the approaching (see step **S120**) and the docking (see step **S150**) of the vehicle **20** to the passenger boarding/alighting platform **30**. The docking of the wheel **45** to the cutout **40** enables the vehicle **20** to stop proximate to the passenger boarding/alighting platform **30**. The gap **60** between the vehicle **20** and the vehicle-facing surface **85** of the passenger boarding/alighting platform **30** is therefore reduced.

(23) FIG. 3B shows a second example of the cutout **40**. The vehicle **20** approaches the stopping station **15** by driving sideways in this second example. The wheels **45** are rotated by a 90° steering angle relative to the forward driving direction **FW** of the vehicle **20**. The vehicle **20** can therefore drive sideways for the approaching (see step **S120**) and the docking (see step **S150**) of the vehicle **20** to the passenger boarding/alighting platform **30**. Setting the wheels **45** to the similar or identical angle relative to the vehicle **20** allows the vehicle **20** to approach the passenger boarding/alighting platform **30** by driving sideways. The driving sideways of the vehicle **20** allows a reduction in the

size of the plurality of the vehicle stopping surfaces **90** as the vehicle **20** is capable of maneuvering in a small surface/area. The driving sideways of the vehicle **20** therefore allows the vehicle to approach the passenger boarding/alighting platform **30** in a fast and efficient manner. The driving sideways could also be done by rotating the wheels at angles other than 90° relative to the forward driving direction FW for the approaching of the station **15**. The wheels **45** could also be set to 45° , to 60° , or at other angles relative to the forward driving direction FW enabling the driving sideways of the vehicle **20**.

(24) In this second example, the cutout width **40_w** corresponds substantially to the width of the wheel **45**. In this second example, the cutout depth **40_d** is dimensioned to allow the wheel **45** to dock into the cutout **40** partially or entirely. This docking of the wheels **45** allows the vehicle **20** to stop proximate to the passenger boarding/alighting platform **30**. The gap **60** is therefore also reduced in this second example.

(25) FIG. 3C shows a third example of the cutout **40**. The cutout **40** is dimensioned to be of different shape and size compared to the wheel hull curve **50** in the third example. The cutout **40** of the third example is dimensioned to accommodate the wheel hull curve **50** with an additional space between the wheel **45** and the cutout **40**. The gap **60** between the vehicle **20** and the passenger boarding/alighting platform **30** will be reduced by allowing the wheel **45** to rotate in the cutout **40** when in the docked position.

(26) FIG. 3D shows a fourth example of the cutout **40**. The vehicle **20** approaches the stopping station **15** by driving sideways in a so-called “crab walk” or “dog walk” approach. The crab walk describes independently steering the wheels **45**. The independent steering of the wheels can be achieved by having a four wheel steering system. The independent steering of the wheels **45** allows the wheels **45** to be set to similar or different steering angles relative to the forward driving direction FW and relative to other ones of the wheels **45**. The front wheels **45_f** are usually set to steer in a similar or substantially identical direction as the rear wheels **45_r**. The front wheels **45_f** and the rear wheels **45_r** may also be set to different steering angles for turning of the vehicle. This steering of the wheels **45** allows the vehicle **20** to turn efficiently.

(27) The front wheels **45_f** are set to steer towards the passenger boarding/alighting platform **30** while the rear wheels **45_r** might be set to steer away from the passenger boarding/alighting platform **30** when approaching the passenger boarding/alighting platform **30**. By setting these steering angles for the wheels **45**, the vehicle **20** is able to maneuver swiftly and initiate a turn only requiring a small one of the plurality of vehicle stopping surfaces **90**. In order to approach the passenger boarding/alighting platform **30** more efficiently, the angles of the wheels **45** can be set to change during the approaching of the passenger boarding/alighting platform **30**.

(28) The form or shape of the cutouts **40** is dimensioned to allow the wheels **45** to dock into the cutouts **40** partially or entirely as can be seen from FIGS. 3A to 3D. The wheels **45** on the side of the vehicle **20** facing the passenger boarding/alighting platform **30** and the corresponding wheel hull curve **50** are able to be accommodated within the cutouts **40**. This accommodating of the wheel **45** and/or of the wheel hull curve **50** in the cutouts **40** allows the vehicle **20** to stop proximate to the passenger boarding/alighting platform **30**. The wheels **45** do not interfere with the with the vehicle-facing surface **85** when docked to the cutouts **40**. The docking thereby reduces the gap **60** (see FIG. 3A to 3D). The vehicle **20** therefore does not need to maintain a distance from the vehicle-facing surface **85** of the passenger boarding/alighting platform **30** for the turning of the wheels **45**. The vehicle **20** can, knowing the exact position of the cutouts **40** in the stopping station **15** and using the onboard positioning unit **26**, stop at the passenger boarding/alighting platform **30** and dock the wheels **45** to the cutouts **40**.

(29) FIG. 4 shows a flowchart describing a method for controlling of the vehicle **20** requesting to stop at the stopping station **15** to one of a plurality of the vehicle stopping surfaces **90**. An approaching vehicle **20** requests to stop at the stopping station **15** in step S10. The requesting is done by sending a request using the vehicle antenna **25** and contains, for example, an expected

arrival time of the vehicle **20** at the stopping station **15** or a number of passengers travelling in the vehicle **20**. The stopping station **15** receives the request from the vehicle **20** via the stopping station communication unit **17** in step **S15**.

(30) The stopping station control unit **16** selects, in step **S20**, one of the plurality of the vehicle stopping surfaces **90** for the stopping of the requesting vehicle **20**. The selecting of the vehicle stopping surfaces **90** is done, for example, based on the estimated arrival time of the vehicle **20** or the number of passengers travelling in the vehicle **20**. The selecting may also be based on a predicted availability of the vehicle stopping surfaces **90**. The predicted availability can be calculated by the stopping station control unit **16** knowing the approaching vehicles **20** and the number and position of the vehicle stopping surfaces **90**. In other words, the vehicle stopping surfaces **90** are dynamically allocated to the requesting vehicles **20** based on a demand.

(31) The stopping station control unit **16** knows from, for example, the vehicles **20**, the roadside control unit, or sensors disposed in the transportation network **100** or disposed at the stopping station **15**, a position of the vehicles **20** relative to the stopping station **15**. The stopping station control unit **16** also calculates, knowing the position of the vehicles **20**, the route for the requesting vehicle **20** to the selected vehicle stopping surface **90** and detects, in step **S25**, conflicts between the approaching ones of the vehicles **20**. This calculating of the route to the vehicle stopping surface **90** is done independently from the calculating of the route by the onboard processor **27** of the vehicle **20** (see above). The stopping station control unit **16** knows from this calculation the routes of the plurality of the vehicles **20** approaching the stopping surfaces **90** of the stopping station **15**.

(32) If the stopping station control unit **16** detects, in step **S25** the conflict between ones of the approaching vehicles **20** having requested the stopping at the stopping station **15**, the stopping station control unit **16** determines conflict avoidance instructions for avoidance of the determined conflict in step **S30a**. The selected one of the plurality of the vehicle stopping surfaces **90** and the conflict avoidance instructions are transmitted to the requesting vehicle **20** in step **S35a**. The requesting vehicle **20** receives items of data identifying the selected vehicle stopping surface **90** and the route to the selected vehicle stopping surface **90** in step **S40a**. The vehicle **20** approaches (see also step **S120** in FIG. 6), using the received items of data on the vehicle stopping surface **90** and the conflict avoidance instructions and stops (see step **S140** in FIGS. 6 and **S220** in FIG. 6) the vehicle stopping surface **90** in step **S45a**.

(33) If the stopping station control unit **16** does not detect the conflict between ones of the approaching vehicles **20** having requested to stop at the stopping station **15** in step **S25**, the selected vehicle stopping surfaces **90** is transmitted to the requesting vehicle **20** in step **S35b**. The requesting vehicle **20** receives items of data identifying the selected vehicle stopping surface **90** in step **S40b**. The vehicle **20** approaches (see also step **S120** in FIG. 6), using the received items of data on the vehicle stopping surface **90** and stops (see step **S140** in FIGS. 6 and **S220** in FIG. 6) on the vehicle stopping surface **90** in step **S45b**.

(34) FIG. 5 shows a flowchart describing a method for controlling a departure of the vehicle **20** requesting to depart from the stopping station **15**. In step **S50** the vehicle requests to depart from the stopping station **15**. The requesting to depart from the stopping station **15** by the vehicle **20** is done by a departure request using the vehicle antenna **25** and the stopping station communication unit **17**. The stopping station control unit **16** receives the departure request to depart from the stopping station **15** by the requesting vehicle **20** in step **S55**. The stopping station control center **16** determines, in step **S60**, a route for the leaving of the stopping station **15** by the requesting vehicle **20**. The determining of the route for the leaving of the stopping station **15** is done using the known position of the vehicles **20** and the known position of the vehicle stopping surfaces **90**. The stopping station communication unit **17** transmits, in step **S65**, the determined route for the leaving of the stopping station **15** to the requesting vehicle **20**. The requesting vehicle **20** leaves, using the route received from the stopping station control center **16**, the stopping station **15** in step **S70**. The

stopping station control unit **16** is capable of monitoring the leaving of the vehicle **20** using the sensors. The stopping station control unit **16** monitors, for example, if the passenger or an obstacle is on the vehicle stopping surface **90** or close to the vehicle stopping surface **90**. The stopping station control unit **16** may inform the vehicles **20** of the passenger or the object to avoid a conflict with the vehicles **20**.

(35) FIG. **6** shows a flowchart describing a method for boarding/alighting of the passengers at the passenger boarding/alighting platform **30** into/from the vehicle **20** in the transportation network **100**. The vehicle **20** knows the approximate position of the vehicle stopping surface **90** in the transportation network **100** from the structure model **75** stored in the vehicle memory **28**. The vehicle **20** detects in step **S100**, using the onboard processor **27** and the onboard positioning unit **26**, the exact position of the vehicle stopping surface **90** disposed in the transportation network **100**. A check is conducted in step **S110** to verify that the vehicle **20** has detected the vehicle stopping surface **90**.

(36) If the vehicle **20** has detected the vehicle stopping surface **90** in step **S100**, the vehicle approaches the vehicle stopping surface **90** in step **S120**. If the vehicle has not detected the stopping station, step **S100** is reiterated. Approaching the one of the plurality of the vehicle stopping surfaces **90** is done, by driving from a road of the transportation network **100** onto the one of the plurality of the vehicle stopping surfaces **90**.

(37) The vehicle **20** aligns, in step **S130**, the wheels **45** of the vehicle **20** to the position of the cutouts **40** disposed on the vehicle-facing surface **85** of the passenger boarding/alighting platform **30**. Aligning the wheels **45** of the vehicle **20** to the cutouts of the vehicle-facing surface **85** of the passenger boarding/alighting platform **30** comprises calculating the alignment of the vehicle **20** with the cutouts **15** in the onboard processing unit **26** of the vehicle **20**. The calculating is done using the known position of the stopping station **15** (see step **S100**) and items of information on the position of the cutouts **40** relative to the stopping station **15** stored in the structure model **75**. Aligning the wheels **45** further comprises calculating the angle of the wheels **45** relative to the vehicle **20** during the approaching of the vehicle **20** and setting the angle of the wheels **45** to the calculated angle during the approaching to the passenger boarding/alighting platform **30**. The cutouts **40** are dimensioned according to the wheel hull curve **50** of the vehicle and enable the rotation of the wheel **45**.

(38) The vehicle **20** stops, in step **S140**, proximate to the passenger boarding/alighting platform **30**. The aligning of the wheels **45** to the cutouts **40** disposed in the passenger boarding/alighting platform **30** enables the stopping of the vehicle **20** proximate to the passenger boarding/alighting platform **30**. In a case, where the wheels **45** are flush or aligned with the side **21** of the vehicle **20**, the dimension of the cutouts **40** is, for example, relatively small since the cutouts **40** might only be needed for the turning of the wheels **45** during the stopping **S140** and the docking **S150** of the vehicle **20** or during the leaving **S170** of the stopping station **15**. In a case, where the wheels **45** stick out from the vehicle **20**, the dimension of the cutouts **40** is, for example, bigger (compared to the dimension of the cutouts when the wheels **45** are flush or aligned with the side **21**).

(39) The vehicle **20** docks the wheels **45** of the vehicle **20** in the cutouts **40** of the passenger boarding/alighting platform **30** in step **S150**. In step **S160**, the passengers board/alight the vehicle **20** from/onto the upper surface **80** of the passenger boarding/alighting platform **30**.

(40) The vehicle **20** leaves the stopping station **15** in step **S170** after the boarding/alighting of the passengers has been completed. The leaving of the stopping station **15** is done in a similar way as the approaching to the stopping station **15** (see step **S120**) above.

(41) FIG. **7** shows a flow chart describing a method for management of the vehicles **20** at the stopping station **15** in the transportation network **100**. The vehicle aligns, in step **S210**, the wheels **45** of the vehicle **20** to the cutouts **40**, as described in step **S130** of FIG. **6** (see above). In step **S220**, the vehicle stops at the stopping station **15** proximate to the passenger boarding/alighting platform **30**. The vehicle **20** docks the wheels **45** in the cutouts **40** in step **S230**.

(42) 15 stopping station **16** stopping station control unit **17** stopping station communication unit **20** vehicle **21** side **25** vehicle antenna **26** onboard positioning unit **27** onboard processor **28** vehicle memory **30** passenger boarding/alighting platform **40** cutout **40w** cutout width **40h** cutout height **40d** cutout depth **45** wheel **45f** front wheel **45r** rear wheel **50** wheel hull curve **75** structure model **80** upper surface **85** vehicle-facing surface **90** vehicle stopping surface **100** transportation network

Claims

1. A stopping station for use in a transportation network, the stopping station comprising: a passenger boarding/alighting platform having an upper surface for boarding of passengers into a vehicle; the passenger boarding/alighting platform having a vehicle-facing surface comprising at least two of a cutout for docking of a wheel, wherein the form of the cutout is dimensioned according to a wheel hull curve of at least one wheel of the vehicle and the number of cutouts disposed on the passenger boarding platform correlates to the number of wheels of the vehicle on a side of the vehicle facing the passenger boarding/alighting platform; and a plurality of vehicle stopping surfaces adjacent to the vehicle-facing surface, wherein the plurality of the vehicle stopping surfaces are used for the stopping of the vehicle during boarding/alighting of the passengers.
2. The stopping station according to claim 1, wherein: the passenger boarding platform is made of at least one of concrete, asphalt, steel, steel-reinforced concrete, lawn stones, or wood.
3. The stopping station according to claim 1, wherein: the cutout is dimensioned to enable rotation of the docked wheel.
4. A method for controlling a movement of one of a plurality of vehicles to one of a plurality of vehicle stopping surfaces at a stopping station, the method comprising: receiving from the one of the plurality of the vehicles a request for stopping at the one of the plurality of the vehicle stopping surfaces of the stopping station; selecting the one of the plurality of the vehicle stopping surfaces for stopping of a requesting one of the vehicles at the one of the plurality of vehicle stopping surfaces; and transmitting the selected vehicle stopping surface to the requesting one of the plurality of the vehicles; wherein the selecting of the one of the plurality of the stopping surfaces comprises selecting cutouts for docking of wheels of the requesting one of the plurality of vehicles to a passenger boarding/alighting platform, and wherein the form of the cutout is dimensioned according to a wheel hull curve of at least one wheel of the vehicle and the number of cutouts disposed on the passenger boarding platform correlates to the number of wheels of the vehicle on a side of the vehicle facing the passenger boarding/alighting platform.
5. The method of claim 4, wherein: the selecting of the one of the plurality of the stopping surfaces is based on at least one of an expected arrival time of the requesting one of the vehicles or a number of passengers travelling in the requesting one of the vehicles.
6. The method of claim 4, wherein: the calculating is done using a structure model stored in the stopping station control unit, a known position of the requesting one of the vehicles in relation to the stopping station, and a position of the cutouts in relation to the stopping station.
7. The method of claim 4, wherein: docking of the wheels to the cutouts disposed on a passenger boarding/alighting platform adjacent to the determined one of the plurality of the vehicle stopping surfaces.
8. The method of claim 4, wherein: the transmitting of the selected vehicle stopping surfaces further comprises transmitting of conflict avoidance instructions.
9. The method of claim 4, further comprising: detecting a conflict between approaching ones of the vehicles.
10. The method of claim 9, further comprising: determining conflict avoidance instructions for avoidance of the conflict between the approaching ones of the vehicles.

11. A method for controlling a departure of vehicles from a stopping station in a transportation network, the method comprising: receiving, from one of a plurality of vehicles docked to a cutout disposed on a passenger boarding/alighting platform adjacent to the one of a plurality of vehicle stopping surfaces, a request for departure from the stopping station; determining, by a stopping station control unit, a route for leaving of the requesting one of the vehicles from the one of the vehicle stopping surfaces; and transmitting the determined route for the leaving of the vehicle stopping surfaces to the requesting one of the plurality of vehicles.

12. The method of claim 11, wherein: the determining of the route for the leaving of the stopping station is done using a known position of a plurality of vehicles docked to the cutouts of the stopping station.

13. A method for boarding/alighting passengers at a passenger boarding/alighting platform in/from a vehicle in a transportation network, the method comprising: approaching a vehicle stopping surface by the vehicle; aligning wheels of the vehicle with a plurality of cutouts disposed in a vehicle-facing surface of the passenger boarding/alighting platform, wherein the form of the cutout is dimensioned according to a wheel hull curve of at least one wheel of the vehicle and the number of cutouts disposed on the passenger boarding platform correlates to the number of wheels of the vehicle on a side of the vehicle facing the passenger boarding/alighting platform; stopping the vehicle proximate to the passenger boarding/alighting platform; docking the wheels of the vehicle in the cutouts of the to the passenger boarding/alighting platform; boarding/alighting of passengers at the vehicle; and leaving the stopping station by the vehicle.

14. The method according to claim 13, wherein: detecting the vehicle stopping surface disposed in the transportation network is done using an onboard positioning unit and an onboard processor of the vehicle.

15. A method for management of vehicles at a stopping station in a transportation network, the method comprising: aligning wheels of the vehicle with a plurality of cutouts disposed at a vehicle-facing surface of the passenger boarding/alighting platform, wherein the form of the cutout is dimensioned according to a wheel hull curve of at least one wheel of the vehicle and the number of cutouts disposed on the passenger boarding platform correlates to the number of wheels of the vehicle on a side of the vehicle facing the passenger boarding/alighting platform; stopping of the vehicle proximate to the passenger boarding/alighting platform; and docking the wheels of the vehicle in the cutouts of the to the passenger boarding/alighting platform.
